Course 35. Object Oriented System Design (Web Course)

Faculty Coordinator(s) :

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Detailed Syllabus :

Section 1: Basic Concepts

Module 1 : Basic Concepts of Object-Orientation:
Data abstraction -Encapsulation -Inheritance -Aggregation
(2 hours)

Module 2 : Object-Oriented Programming
Evolution of OO programming languages -Simula, Smalltalk, Eiffel, C++, Java, C#
(2 hours)

Module 3 : Fundamentals of OOP
Class -Object -Message -Inheritance -Dynamic binding - Polymorphism -Virtual functions -Virtual table construction -Different perspectives on inheritance - Interface V s Implementation inheritance - Single V s Multiple inheritance - Mixins
(6 hours)

Module 4 : Prototype based OO languages:
Class Vs Prototype -Cloning -Delegation -SELF -Object behavioural evolution - Dynamic inheritance
(2 hours)

Module 5: Theory of objects:
Type -Subtype -Substitution principle -Covariance -Contravariance- Overloading -Parametric polymorphism -Inclusion polymorphism - Lambda Calculus -Object Calculus -Algebra -Denotational semantics
(4 hours)

Section 2: Object-Oriented Software Development Lifecycle

Module 6 : Object-Oriented Analysis:
Class-Responsibility-Collaboration (CRC) technique - Coad & Yourdon method - Bailin's Entity Data Flow Diagram (EDFD)
(3 hours)

Module 7: Object-Oriented Design:

Booch Methodology -Object Modeling Technique -Use Case driven approach -Responsibility driven design

(2 hours)

Module 8: Unified Modeling Language (UML):

Class diagram -Object diagram -Sequence diagram -Use case diagram -Colloboration diagram - Statechart diagram -Activity diagram - Component diagram -Deployment Diagram

(5 hours)

Module 9 : Reuse Mechanisms:

Analysis patterns - Design patterns - Coding patterns - Architecture Patterns - Metapatterns

(5 hours)

Module 10 : Object-Oriented Software Testing:

Fundamentals of software testing - Class testing - Integration testing - Use case scenario testing - Inheritance - Regression testing - Stress testing

(5 hours)

Module 11: OO Software LifeCycle and OO metrics

Fountain Model -Chidamber & Kemherer Metrics

(3 hours)

Section 3: Advanced Topics

Module 12 : Distributed Objects:

Fundamentals of distributed systems - Middleware - Common Object Request Broker Architecture (CORBA), Enterprise Java Beans (EJB) Architecture - .NET Common Language Runtime

(6 hours)

Module 13 : Distributed OO Programming Languages:

Distributed shared object -Linda - Orca - Charm++ - Concurrent Eiffel

(4 hours)

Module 14: Frameworks:

Framework Cookbooks – Model – View - Controller (MVC) framework - Patterns related to MVC - Taligent framework

(4 hours)

Module 15: Object-Oriented Operating Systems:

Kernel structuring using OO concepts - MUSE operating system - Reflection

(3 hours)

Module 16: Miscellaneous Topics

Aspect Oriented Programming (AOP), Subject Oriented Programming (SOP), .Object role modeling, Object - Oriented databases

(4 hours)

References:

1. *Object Oriented Programming - An Evolutionary Approach* by Brad. J. Cox
2. *The Object Primer -The Application Developer's Guide to Object Orientation and the UML* by Scott Ambler
3. *Object-Oriented Software Construction* by Bertrand Meyer
4. *Object Oriented Design with Applications* by Grady Booch
5. *Object-Oriented Modeling and Design* by J. Rambaugh et.al.
6. *Design Patterns for Object-Oriented Software Development* by Wolfgang Pree
7. *Design Patterns - Elements of Reusable Object - Oriented Software* by E. Gamma et.al.
8. *Unified Modeling Language Reference Manual* by James Rambaugh et.al.
9. *Theory of Objects* by Luca Cardelli and Martin Abadi
10. *Introduction to Object Oriented Databases* by W. Kim
11. *Annotated C++ Reference Manual* by M.Ellis and B. Stroustrup